Supplementary Information

BrainEVO: A brain inspired evolving deep learning architecture for cross-domain adaptation

1 Proposition and Proof in our paper

1.1 Proposition S.1: Gradient alignment characterizes local cross-domain transfer compatibility

Proposition. Consider two domains $\mathcal{D}_i$ and $\mathcal{D}_j$ with corresponding empirical losses $\mathcal{L}_i(\Theta)$ and $\mathcal{L}_j(\Theta)$ under the same model state Θ. Denote their gradients by,

$$\mathbf{g}_i = \nabla_\Theta \mathcal{L}_i(\Theta), \qquad \mathbf{g}_j = \nabla_\Theta \mathcal{L}_j(\Theta). \tag{1}$$

Assume that $\mathcal{L}_j$ is differentiable and L_j-smooth. After performing one gradient-descent update using domain $\mathcal{D}_i$,

$$\Theta^+ = \Theta - \eta \mathbf{g}_i, \tag{2}$$

the loss variation on domain $\mathcal{D}_j$ satisfies

$$\mathcal{L}_j(\Theta^+) - \mathcal{L}_j(\Theta) \leq -\eta \mathbf{g}_j^\top \mathbf{g}_i + \frac{L_j \eta^2}{2} \|\mathbf{g}_i\|_2^2. \tag{3}$$

For a sufficiently small learning rate η, the first-order variation is therefore

$$\Delta \mathcal{L}_{j \leftarrow i} = \mathcal{L}_j(\Theta^+) - \mathcal{L}_j(\Theta) \approx -\eta \mathbf{g}_j^\top \mathbf{g}_i. \tag{4}$$

Consequently, positive gradient alignment $\mathbf{g}_j^\top \mathbf{g}_i > 0$ indicates locally compatible optimization directions, whereas negative alignment $\mathbf{g}_j^\top \mathbf{g}_i < 0$ indicates potential optimization conflict between the two domains.

Proof. Since $\mathcal{L}_j$ is L_j-smooth, for any parameter perturbation $\Delta\Theta$ we have

$$\begin{aligned} \mathcal{L}_j(\Theta + \Delta\Theta) \leq & \mathcal{L}_j(\Theta) + \langle \nabla_\Theta \mathcal{L}_j(\Theta), \Delta\Theta \rangle \\ & + \frac{L_j}{2} \|\Delta\Theta\|_2^2. \end{aligned} \tag{5}$$

Substituting $\Delta\Theta = -\eta \mathbf{g}_i$ yields

$$\mathcal{L}_j(\Theta^+) \leq \mathcal{L}_j(\Theta) - \eta \langle \mathbf{g}_j, \mathbf{g}_i \rangle + \frac{L_j \eta^2}{2} \|\mathbf{g}_i\|_2^2. \tag{6}$$

Hence,

$$\mathcal{L}_j(\Theta^+) - \mathcal{L}_j(\Theta) \leq -\eta \mathbf{g}_j^\top \mathbf{g}_i + \frac{L_j \eta^2}{2} \|\mathbf{g}_i\|_2^2. \tag{7}$$

When η is sufficiently small, the second-order term is dominated by the first-order term, giving

$$\Delta\mathcal{L}_{j\leftarrow i} \approx -\eta \mathbf{g}_j^\top \mathbf{g}_i. \tag{8}$$

Therefore, when $\mathbf{g}_j^\top \mathbf{g}_i > 0$, an update toward minimizing $\mathcal{L}_i$ also tends to decrease $\mathcal{L}_j$, indicating positive local transfer. In contrast, $\mathbf{g}_j^\top \mathbf{g}_i < 0$ means that the update favored by $\mathcal{D}_i$ tends to increase the loss of $\mathcal{D}_j$, revealing optimization conflict or potential negative transfer. This completes the proof.

1.2 Proposition IV.1: Gradient magnitude characterizes local optimization demand

Proposition

Let $z = (x, y)$ be a training sample and define the sample-wise loss as $\ell_z(\Theta) = \mathcal{L}(f_\Theta(x), y)$. Assume that $\ell_z(\Theta)$ is differentiable and L-smooth with respect to Θ. For a fixed model state Θ and a step size $\eta \in (0, 1/L]$, consider a one-step gradient update

$$\Theta_z^+ = \Theta - \eta \nabla_\Theta \ell_z(\Theta). \tag{9}$$

Then,

$$\left\|\Theta_z^+ - \Theta\right\|_F = \eta \left\|\nabla_\Theta \ell_z(\Theta)\right\|_F, \tag{10}$$

and

$$\ell_z(\Theta) - \ell_z(\Theta_z^+) \geq \frac{\eta}{2} \left\|\nabla_\Theta \ell_z(\Theta)\right\|_F^2. \tag{11}$$

Therefore, under a fixed model state, a larger gradient magnitude indicates that fitting the corresponding sample requires a larger local parameter displacement and is associated with a larger optimization pressure.

Proof of Proposition IV.1

Proof. Let $g_z(\Theta) := \nabla_\Theta \ell_z(\Theta)$ and $\Theta_z^+ = \Theta - \eta g_z(\Theta)$.

We first prove the relation between the gradient norm and the parameter update magnitude. By the definition of one-step gradient descent, $\Theta_z^+ - \Theta = -\eta g_z(\Theta)$. Taking the Frobenius norm on both sides gives,

$$\|\Theta_z^+ - \Theta\|_F = \eta \|g_z(\Theta)\|_F = \eta \|\nabla_\Theta \ell_z(\Theta)\|_F \tag{12}$$

Next, since $\ell_z(\Theta)$ is assumed to be differentiable and L-smooth with respect to Θ, by the standard smoothness inequality, for any Θ' we have

$$\ell_z(\Theta') \leq \ell_z(\Theta) + \langle \nabla_\Theta \ell_z(\Theta), \Theta' - \Theta \rangle + \frac{L}{2} \|\Theta' - \Theta\|_F^2 \tag{13}$$

Substituting $\Theta' = \Theta_z^+ = \Theta - \eta \nabla_\Theta \ell_z(\Theta)$ yields $\ell_z(\Theta_z^+) \leq \ell_z(\Theta) - \eta \|\nabla_\Theta \ell_z(\Theta)\|_F^2 + \frac{L\eta^2}{2} \|\nabla_\Theta \ell_z(\Theta)\|_F^2$. Rearranging terms, we obtain,

$$\ell_z(\Theta) - \ell_z(\Theta_z^+) \geq \left(\eta - \frac{L\eta^2}{2}\right) \|\nabla_\Theta \ell_z(\Theta)\|_F^2 \tag{14}$$

When $\eta \leq 1/L$, we have

$$\eta - \frac{L\eta^2}{2} \geq \eta - \frac{\eta}{2} = \frac{\eta}{2} \tag{15}$$

and therefore

$$\ell_z(\Theta) - \ell_z(\Theta_z^+) \geq \frac{\eta}{2}\|\nabla_\Theta \ell_z(\Theta)\|_F^2 \tag{16}$$

Hence, under a fixed model state Θ, the gradient norm of a sample determines both the magnitude of the induced parameter update and a lower bound on the achievable one-step loss decrease. This result establishes that the gradient magnitude provides a model-conditioned characterization of the local optimization demand induced by a sample. In BrainEVO, this quantity is used only as the micro-level signal for difficulty-aware ordering; the methodological contribution further relies on the directional geometry of gradients for characterizing inter-domain transfer compatibility. □

1.3 Proposition IV.3: Mutual-information increase induced by effective neuron activation

Proposition

Given the learning domain D_k and next domain D_{k+1}, we have the Dropout probability p_k and the L2 regularization coefficient λ_k. We also have the Dropout probability and L2 regularization coefficient at the next learning domain satisfying $\{p_k > p_{k+1},\ \lambda_k > \lambda_{k+1}\}$. And we can denote the informativeness of knowledge encapsulated in model parameters during learning the (k+1)-th domain D_{k+1} as the mutual information between learnable parameters (weights) $\boldsymbol{W}$ and $\mathcal{D}_{1:k+1}$, i.e., $I(\boldsymbol{W}_{k+1}; \mathcal{D}_{1:k+1})$. When learning domain shift from D_k to D_{k+1}, the knowledge within model can be expanded by releasing regularization factors from learning domain $\boldsymbol{k}$ to domain $\boldsymbol{k+1}$, i.e.,

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \gamma(\mathbb{E}[I(\boldsymbol{W}_{k+1}; \mathcal{D}_{1:k+1})] - \mathbb{E}[I(\boldsymbol{W}_k; \mathcal{D}_{1:k})]) > 0 \tag{17}$$

where scaling factor $\gamma > 0$ and $\mathbb{E}[\cdot]$ denotes the expectation of such mutual information as p_k is a probability.

Proof of Proposition IV.5

Proof. Let $\Theta_{\mathcal{A}_k}$ denote the parameters of currently effectively activated neurons, and let $\Theta_{\mathcal{B}}$ denote the newly activated parameters after adapting to a new domain, where $\mathcal{A}_{k+1} = \mathcal{A}_k \cup \mathcal{B}$. Then,

$$\Theta_{\mathcal{A}_{k+1}} = (\Theta_{\mathcal{A}_k}, \Theta_{\mathcal{B}}).$$

According to the chain rule of mutual information,

$$I(\mathcal{D}; \Theta_{\mathcal{A}_{k+1}}) = I(\mathcal{D}; \Theta_{\mathcal{A}_k}) + I(\mathcal{D}; \Theta_{\mathcal{B}} \mid \Theta_{\mathcal{A}_k}). \tag{18}$$

Therefore, the information gain brought by the newly activated parameters is $\Delta I = I(\mathcal{D}; \Theta_{\mathcal{B}} \mid \Theta_{\mathcal{A}_k})$.

We next show that $\Delta I > 0$ when the newly activated neurons receive non-degenerate data-dependent updates. Conditioned on $\Theta_{\mathcal{A}_k}$, their parameter update can be written as,

$$\Theta_{\mathcal{B}} = \Theta_{\mathcal{B}}^0 - \eta G_{\mathcal{B}}(\mathcal{D}, \Theta_{\mathcal{A}_k}) + \xi, \tag{19}$$

where $G_{\mathcal{B}} = \nabla_{\Theta_{\mathcal{B}}} \mathcal{L}(\mathcal{D}; \Theta)$ denotes the data-dependent gradient, and $\xi \sim \mathcal{N}(0, \Sigma_\xi)$ is independent training noise.

Assume that, conditioned on $\Theta_{\mathcal{A}_k}$,

$$G_{\mathcal{B}} \sim \mathcal{N}(\mu_G, \Sigma_G), \qquad \Sigma_G \neq 0.$$

The condition $\Sigma_G \neq 0$ indicates that the newly activated parameters receive non-degenerate updates that vary with the data.

Since the gradient is determined by the training data and the newly activated parameters are updated through the gradient, we have the conditional Markov relation, $\mathcal{D} \to G_{\mathcal{B}} \to \Theta_{\mathcal{B}}$.

Under the above Gaussian update model, the corresponding conditional mutual information is,

$$I(\mathcal{D}; \Theta_{\mathcal{B}} \mid \Theta_{\mathcal{A}_k}) = \frac{1}{2}\mathbb{E}\left[\log\det\left(I + \eta^2 \Sigma_{\xi}^{-1}\Sigma_G\right)\right]. \tag{20}$$

Because $\Sigma_\xi \succ 0$, $\Sigma_G \succeq 0$, and $\Sigma_G \neq 0$, at least one eigenvalue of $\Sigma_{\xi}^{-1}\Sigma_G$ is positive. Therefore, $I(\mathcal{D}; \Theta_{\mathcal{B}} \mid \Theta_{\mathcal{A}_k}) > 0$.

Consequently, we have $I(\mathcal{D}; \Theta_{\mathcal{A}_{k+1}}) > I(\mathcal{D}; \Theta_{\mathcal{A}_k})$, showing that effectively activating new neurons with non-redundant data-dependent updates increases the information encoded in the model.

Furthermore, let $\Theta_{\mathcal{B}} = (\theta_1, \ldots, \theta_m)$ contain m newly activated neurons. By repeatedly applying the chain rule,

$$\Delta I = \sum_{r=1}^{m} I(\mathcal{D}; \theta_r \mid \Theta_{\mathcal{A}_k}, \theta_1, \ldots, \theta_{r-1}). \tag{21}$$

If each newly activated neuron contributes at least $\delta > 0$ conditional non-redundant information, i.e.,

$$I(\mathcal{D}; \theta_r \mid \Theta_{\mathcal{A}_k}, \theta_1, \ldots, \theta_{r-1}) \geq \delta, \tag{22}$$

then $\Delta I \geq m\delta$. □

Hence, under the non-redundancy condition, the information acquired by the model increases at least linearly with the number of effectively activated neurons. This completes the proof.

2 Related work review

Learning for cross-domain adaptation. A central challenge in cross-domain adaptation lies in that distributions differ between training and testing sets [25]. Cross-domain adaptation can be divided into two aspects, the data distribution changes across space or temporal evolution for the same task, and the domain shift from source to target [66]. Early researches propose to counter temporal distribution shifts, such as traffic dynamics across years, with experience-replay of continual learning [10] and invariant learning [75] for generalization. Actually, the layout or structure of spatial domain is not always static, thus discovering the invariances on graphs including both macro networks [32] and micro biomolecules [60] also becomes a popular research branch with neuro-inspired mechanisms [39]. Specifically, urban environments are evolving on both spatial and temporal aspects [52, 69], thus dynamic explore-to-extrapolate risk objective [52] are constructed to overcome spatial shifts, while decoupled common knowledge and gradient discrepancy [27, 41, 73] are constructed to link the training and testing data with cross-domain samples. For the source-target domain shift, recent literature construct the fast-low learning systems to leverage the commonality and individual pattern during domain shifts [3, 57]. More recently, trials on cross-task learning [66, 70] establish

the correlations among different tasks within similar context. Despite progress, existing methods rarely exploit shared information across inputs and quantify inter-domain relationships, so existing cross-domain adaptation lacks a unified mechanism that can be generalized into both macro and micro worlds, and enables per-domain adaptive selection within a single framework.

Evolutionary algorithm and continual learning. Evolutionary algorithms and continual deep learning are two major paradigms for adaptive models. Inspired by survival of the fittest, evolutionary computation provides a population-based meta-heuristic framework for optimization. Representative methods, such as particle swarm optimization (PSO) [17] and ant colony optimization (ACO) [28], evolve candidate solutions through stochastic search strategies resembling collective biological behaviors. However, they mainly operate under predefined optimization rules and passively explore the solution space, rather than explicitly adapting model parameters and structures to sequentially arriving data or tasks.

To enable data-adaptive evolution, surge of continual learning researches are proposed to ensure a stability-plasticity trade-off in adapting new domains or tasks with data arrival [58]. Continual learning can be summarized into four aspects. 1) Regularization-based methods [1, 26, 46] add additional terms to balance old-new data patterns, but often weaken the acquisition of new knowledge. 2) Replay-based methods exploit the gradient for experience replay [50] while generative replay replaces old samples with generated pseudo-samples, which is particularly suitable for continual zero-shot learning [18]. But they still potentially derives the mismatch between replayed and current distributions. 3) Optimization based solutions construct diverse representation or meta-learning strategies to enable model evolution [22], but lack explicit control over where to preserve and where to evolve. 4) Architecture-based methods enlarge capacity through three perspectives. **i) Progressive neural networks.** Progressive Neural Networks [48] address forgetting by allocating a new sub-network for each new task while freezing previously learned parameters. Their key advantage is strong task isolation, which effectively preserves old knowledge. However, they require continuous parameter growth as tasks accumulate and usually assume clear task boundaries to decide when a new branch should be added. **ii) Adapter-based continual learning.** Adapter-based continual learning [53] keeps a shared backbone and introduces only lightweight task-specific modules, such as adapters, prompts, or LoRA layers. This design is more parameter-efficient than progressive networks and preserves the generalization ability of the shared backbone. However, when task heterogeneity is large, a single shared backbone may be insufficient, and additional mechanisms are often needed to select or compose the appropriate adapters, which may introduce instability [59]. **iii) Routing-based and expert-based methods.** Routing- and expert-based continual learning methods [40, 68] maintain multiple experts and use a router/gating mechanism to decide which expert(s) should process each input. Their main advantage is that they combine parameter sharing with input-dependent specialization. However, routing itself is difficult to optimize and may suffer from expert imbalance, routing collapse, or task interference, which can also result in instability of learning process. Overall, these methods indeed extend continual learning beyond fixed-capacity models. But at the same time, in the specific problem regarding domain data expansion and model, these solutions still face unresolved issues, including unbounded parameter growth, the complexity of explicit task-to-module scheduling, as well as unstable routing behavior.

In contrast, BrainEVO does not rely on adding many extra sub-networks. Instead, it performs adaptive model evolution within the existing architecture through self-organized data ordering, learnable network partition coordination with common-pattern enrichment, and LTP/LTD-inspired memory-learning interaction. This allows the model to evolve with a stable parameter scale and a more stable learning process, which is especially important when new-domain data are limited.

Learning from biological brain. Modern artificial intelligence is deeply inspired by brain structures and functions, from early neural models [47] to recent brain-inspired continual learning studies [3]. The biological brain provides important clues for adaptive intelligence from three aspects. **1) Hierarchical and inter-connected organization** [5, 23, 37]: the cerebral cortex is hierarchically organized and densely interconnected [2, 23], where lower regions process sensory details and higher regions form abstract representations, inspiring multi-level feature abstraction and inter-layer coordination in deep models [16]. **2) Complementary regional functions** [31]: the hippocampus rapidly encodes episodic experiences, while the neocortex gradually extracts stable regularities [6, 42], offering a biological basis for balancing fast adaptation and long-term memorization [34, 44, 74]. **3) Reciprocal interaction at neuronal and circuit levels**: neurotransmitter-based signaling supports synaptic strengthening and weakening, such as long-term potentiation (LTP) and long-term depression (LTD) [7, 29, 38], while circuit-level learning gradually consolidates experience into stable and generalizable knowledge [30, 31, 49, 54, 71]. Inspired by these mechanisms, episodic and working memory have been incorporated into deep models to guide plasticity and integrate new experiences into existing knowledge structures [8, 19]. Nevertheless, as highlighted by *Nature Machine Intelligence* [9], the integration of AI and neuroscience remains largely open. In cross-domain adaptation, key challenges still include how to operationalize easy-to-hard curricula and how to mimic brain-like learning progress and knowledge interaction for reliable pattern transfer [13].

3 Additional experiments

3.1 Detailed analysis of model-conditioned gradient fields and hierarchical curriculum

In this subsection, we provide additional empirical analyses of the model-conditioned gradient field used by BrainEVO. Rather than considering gradient norm alone as the methodological contribution, we examine whether gradients extracted from heterogeneous domains provide informative signals for both domain-wise optimization demand and inter-domain organization. We further evaluate the computational cost and robustness of using such gradient information to construct the proposed hierarchical curriculum.

Visualization of model-conditioned domain gradients

To better understand whether the extracted gradients provide informative model-conditioned representations of different domains, we visualize the corresponding gradient patterns in Fig. 1. For each dataset, gradients are extracted after a short probing stage, with three epochs for spatiotemporal datasets and five epochs for molecular datasets.

The purpose of this visualization is not merely to compare gradient magnitudes. Instead, it illustrates that different domains induce distinguishable parameter-update patterns under the same model, providing empirical motivation for exploiting both gradient magnitude and direction in hierarchical domain organization. In particular, the gradient magnitude reflects the amount of local adaptation required by a domain, whereas differences in gradient directions capture heterogeneous update preferences across domains.

Complexity and efficiency of curriculum learning process

Our hierarchical curriculum learning provides a data organization strategy to deal with disordered data, which is a one-off domain-level pre-processing rather than a repeated cost during every update. To offer

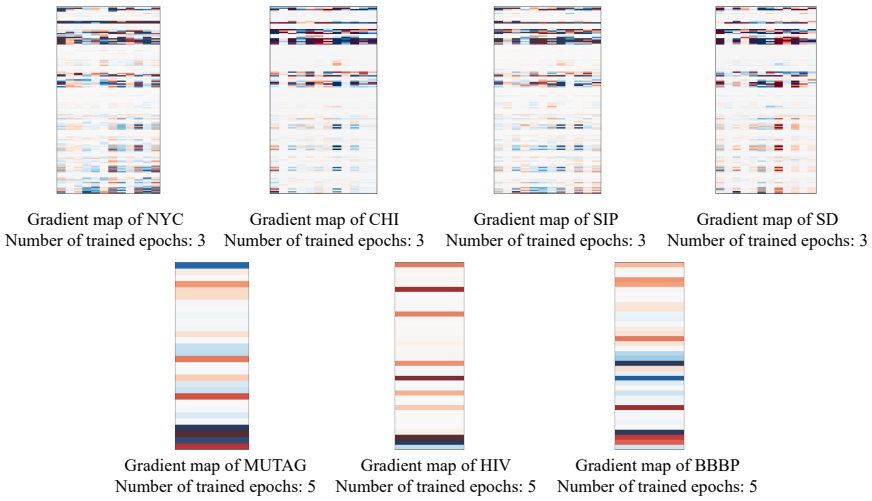


Figure 1: Visualization of model-conditioned gradient patterns across spatiotemporal and molecular domains after lightweight probing. Different domains induce distinguishable gradient magnitudes and directions, which are respectively exploited for optimization-demand estimation and transfer-compatibility modeling in BrainEVO.

Table 1: The runtime statistics for curriculum learning

Dataset	NYC	CHI	SIP	SD	MUTAG	HIV	BBBP
Number of divided domains	16	12	8	6	4	4	4
Runtime ratio of CL (%)	10.81	10.65	9.75	9.34	17.58	12.69	16.82

further evaluation, we first show the complexity of curriculum learning and then provide the evidence of tradeoff between effectiveness and efficiency, regarding performances, number of learning samples, as well as dimensions of gradient vectors.

First, **the runtime ratios (time cost proportion)** between implementing curriculum learning and full learning process for each dataset are shown in Tab. 1, which illustrates the number of divided domains for reference. The results explicitly show that the proportion of curriculum learning is acceptable to overall training process.

Second, the construction of the model-conditioned gradient field does not require full-scale optimization over every domain. Inspired by stochastic gradient estimation, we construct a lightweight probing variant by either estimating domain gradients from a subset of training samples or compressing the gradient representation before compatibility analysis. Specifically, we use 40%, 50%, 60%, and 70% of training samples for gradient estimation, or retain the same proportions of gradient dimensions for subsequent domain organization. Here we present the performances vs computational burden (runtime ratio, %) on NYC for verifying such effectiveness-efficiency tradeoff in Tab. 2.

Result summary. Overall, constructing the gradient field accounts for only 9.34%–17.58% of the full training time across the evaluated datasets. Moreover, substantially reducing either the number of probing samples or the gradient dimensionality results in only minor performance degradation. These results indicate that the proposed gradient-field-based domain organization does not depend on expensive full-data gradient computation and admits an effective accuracy–efficiency trade-off.

Table 2: Performance evaluation with reduced sample-level computational complexity on NYC

Prop. of training samples	40%	50%	60%	70%	100%
Runtime ratio of CL (%)	4.92	5.76	6.74	7.99	10.81
Performance (MAE)	6.559	6.555	6.532	6.503	6.496
Prop. of gradient vector dimension	40%	50%	60%	70%	100%
Runtime ratio of CL (%)	4.86	5.91	6.68	7.92	10.81
Performance (MAE)	6.558	6.548	6.542	6.499	6.496

Discussion on robustness of model-conditioned gradient estimation

The model-conditioned gradient field may vary with random initialization, mini-batch sampling, and training stage. Therefore, BrainEVO does not conceptually rely on a single stochastic gradient snapshot. A more robust implementation can estimate both magnitude and directional information over multiple probing batches or early-stage checkpoints.

For domain $\mathcal{D}_k$, let $\{\mathbf{g}_k^{(r)}\}_{r=1}^{R}$ denote gradient observations obtained from R probing runs. The optimization demand can be estimated using a robust aggregation of their magnitudes,

$$\hat{m}_k = \mathrm{Median}_r \left(\|\mathbf{g}_k^{(r)}\|_2 \right), \tag{23}$$

while the domain update direction can be estimated by,

$$\hat{\mathbf{u}}_k = \frac{\sum_{r=1}^{R} \mathbf{g}_k^{(r)} / (\|\mathbf{g}_k^{(r)}\|_2 + \epsilon)}{\left\| \sum_{r=1}^{R} \mathbf{g}_k^{(r)} / (\|\mathbf{g}_k^{(r)}\|_2 + \epsilon) \right\|_2 + \epsilon}. \tag{24}$$

Such aggregation reduces sensitivity to individual stochastic observations while preserving the two complementary signals required by BrainEVO: optimization demand and inter-domain transfer compatibility. In practice, our results with reduced probing samples and compressed gradient representations further demonstrate that the hierarchical organization remains stable under approximate gradient estimation.

Insights of effectiveness-efficiency trade-off for gradient estimation

In addition, when the number of domains becomes large, full gradient probing may introduce non-negligible computational overhead. In such cases, we can further adopt the coarse gradient estimation strategy evaluated in Supplementary Table. 2, where only a subset of samples is used for gradient probing to achieve a better effectiveness–efficiency trade-off. Actually, aggressive subsampling slightly degrades the final prediction performance, indicating that computational efficiency should be balanced against the accuracy of gradient estimation. To provide a practical scaling rule, we introduce a simple domain-aware sampling heuristic. Let n denote the number of domains and $r(n)$ the proportion of samples used for gradient computation:

$$r(n) = \max \left(r_{\min}, \min \left(1, \frac{n_0}{n} \right) \right),$$

where n_0 denotes the reference number of domains for full gradient probing and $r_{\min}$ specifies the minimum sampling ratio. In practice, we set $n_0 = 16$ and $r_{\min} = 0.4$. Therefore, full gradient computation is retained when the number of domains is moderate, while the sampling ratio gradually decreases as the number of

domains increases, with a lower bound of 40%.

3.2 Baseline, evaluation settings and results

In this section, we elaborate the settings of baselines and introduce one more backbones for both research fields of urban science and biological science, to further demonstrate the fairness of evaluation process, superiority of learning strategy and scalability of BrainEVO. We exploit comprehensive baselines with various training strategies to evaluate the effectiveness of our BrainEVO.

Baselines and metrics for spatiotemporal urban datasets. We employ three standard performance metrics, i.e., Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). Furthermore, our study incorporates 11 baseline models for comparative assessment including 6 STGNNs with different backbones and learning manners, an RNN-based model, 3 attention-based models and a NeuroAI model. We especially take GWN and CMuST as backbones for BrainEVO on spatiotemporal datasets and take GIN and GAT for BrainEVO on molecule datasets.

- *STGCN*: a spatial-temporal aware graph learning method for POI recommendation [67] (STGNN).
- *STGODE*: Spatial-Temporal Graph ODE Networks for Traffic Flow Forecasting [15] (STGNN).
- *GWN*: A graph-based traffic prediction model that integrates TCNs and GCNs [62] (STGNN).
- *GWN-FR*: Train GWN in a full randomly sampling manner [62] (STGNN).
- *GWN-w/o Com*: Train GWN directly on testing dataset [62] (STGNN).
- *STGP*: A prompt-based framework capable of adapting to diverse tasks in a data-scarce domain [24] (STGNN and multi-source based).
- *AGCRN*: Adaptive graph convolution recurrent network for traffic forecasting [4] (RNN-based model).
- *STTN*: Spatial-temporal transformer networks for traffic flow forecasting [64] (Attention-based model).
- *ASTGCN*: Attention based spatial-temporal graph convolution networks for traffic flow forecasting [21] (Attention-based model).
- *CMuST*: A continuous multi-task spatio-temporal learning frame-work [66] (Attention-based model and multi-source model).
- *PNN*: Progressive Neural Network preserves previous task-specific columns and introduce new columns with lateral connections for knowledge transfer [48] (Continual learning).
- *Hide-PET*: It preserves the pretrained backbone and adapts to new tasks through parameter-efficient components and hierarchical objectives [56](Continual learning).
- *SynEVO*: A neuro-inspired spatiotemporal framework for cross-domain adaptation [35] (NeuroAI model).

Baselines and metrics for biological molecular datasets. For these molecule property modeling, they are mostly classification tasks and we employ three standard metrics, Accuracy, ROC-AUC and F1-score. And we incorporates six baseline models for comparative assessment.

- *GAT*: A graph neural network model that employs attention mechanisms to aggregate information from neighboring nodes [51].

- *GCN*: A graph neural network model that performs neighborhood aggregation through spectral-based graph convolutions [72].
- *GIN*: A graph neural model that uses injective aggregation to achieve powerful graph representation and discrimination [63].
- *GIN-FR*: Train GIN with full randomly sample orders [63].
- *GIN-w/o Com*: Directly train GIN on targeting samples without considering cross-domain commonality [63].
- *PNA (GIN)*: A graph neural model that integrates multi-aggregator and degree-scaled mechanisms for expressive neighborhood representation [11].
- *EMoNet (GIN)*: An environment causal learning for molecule ood generalization [33].
- *NeGo (GIN)*: A negative inference graph OOD framework [65].
- *TMM (GIN)*: A brain-inspired continual learning method that integrates complementary memory systems to mitigate catastrophic forgetting with GIN as backbone for molecule tasks [55].
- *Hide-PET (GIN)*: It preserves the pretrained backbone and adapts to new tasks through parameter-efficient components and hierarchical objectives [56].

Implementation details.

To facilitate the reproducibility of BrainEVO, we provide more details of training and implementations. The datasets were partitioned into training, validation, and testing sets with a ratio of 7:1:2. We follow the general continual learning protocol [58] and allow source domains to be trained and leave target domains as evaluation domains for inference, where targeted samples cannot be accessible during training process. According to the characteristics of datasets themselves, we take the NYC, CHI, and SIP datasets for both cross-source and cross-temporal domain validation, and take SD set for cross-temporal domain evaluation due to its large time span with single attribute.

Learning across domains on urban datasets. For cross-source domain evaluation on the first three spatiotemporal datasets (NYC, CHI, SIP), we respectively reserve TaxiPick, TaxiDrop, and Speed as the target evaluation domains within their datasets, and iteratively training on the remaining data sources. For cross-temporal evaluation on these datasets, each day was divided into four equal intervals, with the final intervals across all source domains held out for assessment. For cross-temporal evaluation on SD dataset, each day was segmented into six equal periods, and the last period was reserved to evaluate temporal domain adaptation.

Learning across domains on molecule datasets. Biological molecules consist of invariant structures like scaffolds and pharmacophores, as well as environment sensitive elements where different substructures imply various properties. To this end, we partition molecules into four domains by whether the molecule has carbonyl group or not and whether it has halogen or not, as these functionalities induce systematic shifts in physicochemical properties, i.e., halogens enhance lipophilicity and stability, while carbonyl groups are highly polar and prone to nucleophilic addition. This yields chemically meaningful domain differences, and learning across these specific domains verifies whether our evolution learning can separate invariant representations from environment-dependent mechanisms. Considering that those molecules with neither carbonyl group

Table 3: Performance comparison on cross-source adaptation on NYC, CHI, SIP and cross-temporal adaptation on SD. Best results are **bold** and the second best results are underlined.

Methods	NYC			CHI			SIP			SD		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
STGCN	6.774	15.853	0.374	1.518	2.804	0.415	0.753	1.492	0.219	12.477	22.055	0.182
STGODE	9.522	22.555	0.481	1.543	2.792	0.433	0.732	1.460	0.211	12.300	20.808	0.180
GWN	10.263	24.535	0.546	1.520	2.873	0.421	0.737	1.473	0.212	18.890	29.220	0.230
GWN-FR	10.076	24.128	0.537	1.522	2.870	0.419	0.738	1.472	0.212	17.661	28.987	0.226
GWN-w/o Com	8.963	21.435	0.489	1.603	2.918	0.421	0.747	1.484	0.217	14.792	25.583	0.190
STTN	7.962	19.544	0.435	1.494	2.699	0.409	0.732	1.461	0.212	13.092	22.054	0.183
AGCRN	8.254	19.301	0.488	1.543	2.805	0.427	0.743	1.469	0.215	12.225	22.094	0.179
ASTGCN	10.323	25.070	0.519	1.536	2.809	0.413	0.743	1.473	0.215	13.079	22.047	0.185
CMuST	6.576	14.954	0.459	1.498	2.781	0.388	0.737	1.497	0.219	10.940	19.113	0.162
SynEVO	6.494	14.885	**0.358**	1.486	2.733	0.361	0.697	1.390	0.205	10.984	18.654	0.157
STGP	6.497	14.882	0.373	1.482	2.678	0.364	0.695	1.392	0.202	17.906	28.651	0.188
PNN	7.430	17.913	0.415	1.528	2.796	0.411	0.735	1.474	0.213	15.663	21.748	0.181
HiDe-PET	7.425	17.836	0.410	1.530	2.788	0.408	0.706	1.412	0.212	16.924	22.651	0.189
BrainEVO (GWN)	6.496	14.422	0.367	**1.453**	2.682	0.364	0.680	1.350	0.198	10.585	17.839	0.143
BrainEVO (CMuST)	**6.492**	**14.421**	0.362	**1.373**	**2.484**	**0.350**	**0.661**	**1.321**	**0.196**	**9.467**	**17.528**	**0.141**

Table 4: Performance comparison on cross-domain adaptation on biological molecular datasets. Best results are **bold** and the second best results are underlined.

Methods	MUTAG			HIV			BBBP		
	ACC(%)	AUC(%)	F1-score(%)	ACC(%)	AUC(%)	F1-score(%)	ACC(%)	AUC(%)	F1-score(%)
GAT	76.4 ± 4.7	79.2 ± 3.1	76.1 ± 4.3	70.9 ± 2.3	72.5 ± 3.0	79.9 ± 4.1	65.4 ± 2.6	69.8 ± 3.4	72.9 ± 3.3
GCN	83.6 ± 5.3	87.6 ± 5.4	82.0 ± 3.7	75.1 ± 1.7	76.3 ± 2.1	82.5 ± 3.7	65.0 ± 2.1	68.7 ± 2.7	70.3 ± 3.6
GIN	89.3 ± 5.2	90.2 ± 4.9	86.6 ± 3.9	75.6 ± 1.4	77.2 ± 1.2	82.6 ± 2.3	66.4 ± 1.9	70.1 ± 2.1	71.7 ± 2.6
GIN-FR	88.6 ± 4.9	90.4 ± 4.7	86.9 ± 4.1	75.8 ± 1.5	77.3 ± 1.6	82.5 ± 2.1	66.9 ± 1.8	70.0 ± 1.7	71.7 ± 2.8
GIN-w/o Com	92.9 ± 3.1	95.7 ± 2.2	90.5 ± 1.3	77.3 ± 1.1	79.9 ± 1.5	84.6 ± 1.4	71.2 ± 1.6	74.9 ± 1.0	77.0 ± 1.1
PNA (GIN)	86.4 ± 4.2	88.4 ± 3.5	82.4 ± 4.0	77.5 ± 1.8	80.1 ± 1.1	84.1 ± 0.9	71.7 ± 1.9	74.7 ± 1.7	76.8 ± 1.7
EMoNet (GIN)	96.4 ± 1.5	96.1 ± 2.3	91.2 ± 1.4	77.8 ± 1.6	80.3 ± 1.5	84.9 ± 1.1	72.1 ± 1.1	75.8 ± 0.8	77.2 ± 1.3
NeGo (GIN)	92.9 ± 1.7	95.5 ± 1.1	90.1 ± 1.6	76.4 ± 1.4	79.7 ± 0.9	84.2 ± 2.2	70.9 ± 0.9	73.9 ± 1.2	76.5 ± 1.1
TMN (GIN)	95.3± 1.4	96.2± 2.7	90.7 ± 1.9	78.5± 1.6	80.6 ± 2.1	84.1 ± 1.6	71.8 ± 2.0	75.4 ± 2.4	77.0 ± 2.3
HiDe-PET (GIN)	94.9 ± 1.5	95.8 ± 1.9	90.7 ± 1.10	78.3 ± 1.4	80.9 ± 2.0	83.6 ± 1.8	71.5 ± 2.7	75.9 ± 1.3	76.5 ± 2.2
BrainEVO (GIN)	**97.9 ± 1.3**	**99.3 ± 0.9**	**91.5 ± 1.2**	**81.5 ± 1.0**	**82.3 ± 1.6**	**86.3 ± 1.7**	**75.1 ± 1.7**	**78.3 ± 1.3**	**79.0 ± 1.6**
BrainEVO (GAT)	88.6 ± 1.5	93.8 ± 1.0	86.7 ± 1.9	76.8 ± 1.0	77.2 ± 1.5	82.4 ± 1.9	73.9 ± 2.0	77.6 ± 1.5	79.8 ± 1.5

nor halogen are more difficult to predict the chemical properties because of lacking the important mentioned groups, we leave the domain which contains molecules with neither carbonyl group nor halogen as the target evaluation domain and utilize other domains as training domains.

Result comparisons.

Comparisons on urban spatiotemporal datasets. The comparison results are reported in Table. 3, which presents the performance of cross-source adaptation on the NYC, CHI, and SIP datasets, as well as cross-temporal adaptation on the SD dataset. In general, our BrainEVO consistently outperforms all baselines across most metrics, achieving either the best or the second-best performance on all four datasets. The improvement beyond best baselines can be found in the last line of Table. 3. Compared with existing baselines on cross-source adaptation, BrainEVO achieves the best overall results. To be specific, BrainEVO reduces prediction errors by averagely 18.8% on NYC, 4.2% on CHI, 7.3% on SIP against all baselines in terms of MAE, which confirms the generalization performance of our method. It's worth noting that BrainEVO reduces MAE by about 24.9% on NYC, 4.7% on CHI, and 8.1% on SIP compared to non-evolving baselines, which confirms the significance of evolving capacity of deep model to deal with domain adaptation. On the cross-temporal dataset SD, BrainEVO sets a new state of the art as well, improving upon baselines by 16.5% on MAE. This demonstrates that BrainEVO can robustly capture temporal dependencies and maintain

stable generalization under non-stationary time dynamics. We also altered the training strategy, including using a fully random sample order and training only on the target domain, to validate the importance of our proposed sequential learning and common-pattern learning mechanisms. And it should be emphasized that BrainEVO consistently improve the performances against SynEVO, confirming the effectiveness on domain transfer when incorporating hierarchical learning and learning-memory interaction mechanism.

Besides, concentrating on the comparison with backbone GWN, BrainEVO significantly outperforms GWN by large margins (about 36.7% on NYC, 4.4% on CHI, 7.7% on SIP and 44.0% on SD in terms of MAE), further verifying that the adaptive evolving scheme has an edge on helping the backbone model jump out of local optimal, which boosts transferability and generalization ability in challenging cross-domain spatiotemporal prediction tasks.

Moreover, by incorporating our BrainEVO with two respective backbones, GWN and CMuST for spatiotemporal datasets while GIN and GAT for biological molecule datasets, we can see that our BrainEVO can seamlessly couple with other spatiotemporal learning architecture and graph-based learning module, and achieve competitive performances. In addition, BrainEVO+CMuST outperforms BrainEVO+GWN but sacrifices more computational costs [35]. Overall, based on above results, we can conclude that our backbone coupling solution still makes sense with both effectiveness and efficiency, and remain these couplings in our main text.

Comparisons on biological molecular datasets The comparison results on micro biological molecular datasets are reported in Table. 4, illustrating performance of functional group-level cross-domain adaptation on MUTAG, HIV and BBBP datasets. Overall, BrainEVO achieves the best accuracy on domain transfer tasks regarding all datasets, demonstrating consistent advantages over existing graph-based baselines. In particular, BrainEVO improves accuracy by about on average 12.5% on MUTAG, 8.1% on HIV, 9.7% on BBBP compared to all baselines, especially outperforms second best EMoNet by about 1.6% on MUTAG, 4.8% on HIV and 4.2% on BBBP, which confirmed the adaptation capability of our method for dealing with biochemical distribution shifts. These results demonstrate that BrainEVO can effectively transfer common patterns between molecules, while maintaining robustness across diverse datasets. Moreover, when comparing with backbone GIN, BrainEVO outperforms GIN by large margins, i.e., about 10.2% on average across all biological molecular datasets in terms of ACC. And we also alter the backbone to GAT, and find that BrainEVO+GAT outperforms most baselines but inferior to BrainEVO+GIN. Thus BrainEVO+GIN becomes the gold combination and all these results can further demonstrate that BrainEVO improves transferability and generalization in cross-domain molecular property prediction tasks.

Comparison of zero-shot adaptation performances on spatiotemporal datasets and molecular tasks To compare our BrainEVO with peer models with zero-shot capacity, we followed the zero-shot learning protocol and carefully perform the fine-tune process on two well-selected zero-shot baselines, ST-GFSL [36] and STMetaNet [43], then providing performances on urban spatiotemporal datasets with standard deviations in Table. 5. We adjusted the important hyperparameters in their neural architecture and training process, i.e., learning rate from {1e-5, 1e-4, 1e-3, 1e-2, 1e-1}, layer depth from {1, 2, 3, 4}, hidden dimension from {8, 16, 32, 64}, and then figured out the best parameters for inference, as shown in Figure. 2. As the best results and the standard deviations via statistical repeated learning-inference demonstrated, we can observe that our BrainEVO still outperforms these two baselines with more robustness. We hypothesize that the weaker performance of the two additional baselines stems from the following factors. (1) ST-GFSL constructs support episodes in the source domain to simulate adaptation to new domains [36], whereas our zero-shot setting provides no access to target-domain data, limiting the applicability of its adap-

Table 5: Comparison of zero-shot cross-temporal adaptation performance on spatiotemporal datasets and cross-domain adaptation performance on molecule datasets.

Methods	NYC	CHI	SIP	SD	MUTAG	HIV	BBBP
	MAE	MAE	MAE	MAE	ACC	ACC	ACC
BrainEVO	13.07	1.96	0.77	16.73	0.64	0.73	0.64
STMetaNet	29.75	3.24	3.95	60.21	/	/	/
ST-GFSL	34.18	5.17	5.76	63.83	/	/	/
Backbone	17.09	4.11	1.04	21.45	0.43	0.36	0.58

tation mechanism. (2) STMetaNet relies heavily on rich external attributes, such as geographic and spatial information [43]. Moreover, neither model explicitly incorporates mechanisms for stable training and flexible expansion of learning capacity to support inference in unseen domains. Under the same conditions, with limited spatiotemporal information available for each city, BrainEVO achieves better performance.

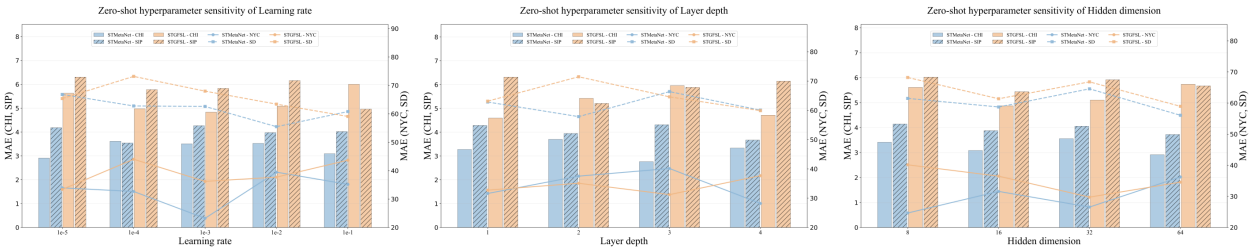


Figure 2: Hyperparameter fine-tuning process of baselines on zero-shot experiments

3.3 Detailed analysis of learning on molecule tasks

OOD degeneration issue on molecular tasks

To further verify that distribution shift indeed causes severe degradation, we additionally report results under a standard random train–test split, where all sample components are mixed and 30% of the data are randomly held out for testing, without using our composition-based grouping, i.e., training and testing on the same domain. The results are demonstrated in Tab. 8. We especially take the averaged OOD results (averaging the performances of four settings "++/+−/−+/−−" of original manuscript Tab.V) as the comparison on the last line of Tab. 8. Under this conventional setting, the performance on HIV and BBBP is much higher than our deliberately constructed OOD settings (92.5% vs 81.8% on HIV and 87.5% vs 75.3%). This confirms that the severe degradation is real and is caused by the simulated distribution shift induced by our sample-group partition protocol.

Therefore, the relatively stable performance of some baselines across the ++ and −− subsets does not imply that the domain shift is weak. Rather, it indicates that all compared methods are being evaluated under the same challenging non-i.i.d. protocol, while the additional comparison with standard random splitting confirms that the constructed setting is indeed much harder and leads to clear performance degeneration.

Domain partition rationale and empirical evidence for molecular dataset

To systematically examine the validity of the molecular-domain construction, we analyze it from three complementary perspectives: chemical interpretability, measurable structural distribution shift, and statistical reliability under limited samples. The partition is grounded in chemically meaningful functional-group information, while Morgan fingerprint analysis confirms that different domains exhibit distinguishable structural patterns rather than arbitrary data separation. In addition, the relatively small MUTAG domains naturally form a controlled low-resource setting, and repeated resampling with statistical testing further verifies that the observed performance gains are robust to small-sample variation. Together, these analyses support the use of the functional-group-based partition as an interpretable and reliable setting for cross-domain and out-of-distribution molecular generalization.

Chemical grounding. The molecular domains are constructed using chemically meaningful functional-group information rather than arbitrary data partitioning. MUTAG originates from the QSAR study of Debnath et al. and contains 188 aromatic and heteroaromatic nitro compounds with substantial structural variation. Its mutagenicity has been shown to be associated with multiple structural and physicochemical factors, including hydrophobicity, LUMO energy, and the number of fused aromatic rings [12]. Functional groups have also been widely adopted as chemical priors for molecular representation and property prediction [14, 45, 61]. Consistent with molecular OOD benchmarks such as GOOD [20], which construct distribution shifts using structural factors such as scaffold and molecular size, the functional-group-based partition used here provides a chemically interpretable and controlled structural distribution shift.

Structural distribution differences. The molecules are divided into four domains: D_1 (no carbonyl and no halogen), D_2 (no carbonyl but with halogen), D_3 (with carbonyl but no halogen), and D_4 (with both carbonyl and halogen). For each molecule x_i, a 2048-bit Morgan fingerprint $\mathbf{z}_i = \mathrm{MorganFP}(x_i; r = 2)$ is computed, and the molecular fingerprints within each domain are averaged to obtain the domain-level representation,

$$\bar{\mathbf{z}}_k = \frac{1}{|D_k|} \sum_{x_i \in D_k} \mathbf{z}_i.$$

The structural similarity between two domains is then measured using the Tanimoto similarity between their domain-level representations.

As shown in Table 6, the self-similarity of each domain representation is 1.0, whereas the similarities between different domains are substantially lower. Specifically, the cross-domain similarities range from 0.5482 to 0.8004 on BBBP, from 0.6830 to 0.8761 on HIV, and from 0.4202 to 0.8501 on MUTAG. These differences indicate that the functional-group partition produces distinguishable molecular structural patterns and measurable distribution shifts, thereby providing a controlled setting for evaluating cross-domain and out-of-distribution generalization.

Statistical reliability under low-resource domains. The relatively small domains in MUTAG also provide a controlled low-resource setting, in which the model receives limited amounts of structurally shifted data and is required to acquire new knowledge while retaining previously learned patterns. To examine whether the observed performance is robust to small-sample variation, a repeated resampling-based evaluation is further conducted on the held-out D_1 domain. Ten test subsets are generated by sampling 10 molecules with replacement in each trial. The same resampled subsets are used to evaluate BrainEVO (GIN), EMoNet (GIN), NeGo (GIN), and vanilla GIN, yielding paired performance measurements across the repeated trials. We subsequently perform paired statistical tests between BrainEVO and each baseline. The results can be found in Table. 7, which show that the performance differences between BrainEVO and

Table 6: Pairwise Tanimoto similarity between domain-level Morgan fingerprints on BBBP, HIV, and MUTAG. Diagonal entries denote within-domain self-similarity, while off-diagonal entries quantify cross-domain structural similarity.

BBBP				
	D_1	D_2	D_3	D_4
D_1	1.0000	0.7476	0.7056	0.5964
D_2	0.7476	1.0000	0.5482	0.6627
D_3	0.7056	0.5482	1.0000	0.8004
D_4	0.5964	0.6627	0.8004	1.0000
HIV				
	D_1	D_2	D_3	D_4
D_1	1.0000	0.8533	0.7986	0.6993
D_2	0.8533	1.0000	0.6830	0.7805
D_3	0.7986	0.6830	1.0000	0.8761
D_4	0.6993	0.7805	0.8761	1.0000
MUTAG				
	D_1	D_2	D_3	D_4
D_1	1.0000	0.7067	0.8501	0.4202
D_2	0.7067	1.0000	0.7299	0.5299
D_3	0.8501	0.7299	1.0000	0.4730
D_4	0.4202	0.5299	0.4730	1.0000

Table 7: Accuracy of four models across ten repeated trials.

Model	Performance of repeated trials (Acc(%))										Average performance (± std.)
EMoNet (GIN)	90	90	90	80	90	100	90	90	90	100	91.00 ± 5.68
NeGo (GIN)	100	90	90	80	90	100	80	90	90	90	90.00 ± 6.67
GIN	80	90	90	70	90	100	90	90	80	90	87.00 ± 8.23
BrainEVO (GIN)	100	90	100	80	100	100	90	100	100	100	96.00 ± 6.99

EMoNet, NeGo, and GIN are statistically significant (96% vs $< 91\%$ also $p < 0.05$), indicating that the improvement of BrainEVO is consistently observed across different resampled test sets rather than being dominated by a particular small test split. We have added this statistical robustness analysis to the revised manuscript to further support the reliability of the results on MUTAG.

3.4 Details of ablation study

For ablation study, we have added one metric, Average Improved Performance (AIP), to measure how overall model improves the performance against the module-level ablative variant by computing the averaged decreases on MAPE (for regression tasks), and averaged increases on ACC (for classification tasks) across

Table 8: Learning performances without cross-source out-of-distribution setting (Accuracy reported)

Dataset	MUTAG	HIV	BBBP
GAT (i.i.d.)	0.836	0.779	0.768
GIN (i.i.d.)	0.929	0.896	0.831
BrainEVO (i.i.d.)	0.979	0.925	0.875
BrainEVO (Averaged OOD)	0.980	0.818	0.753

Table 9: Ablation studies of cross-temporal tasks on spatiotemporal sets and cross-domain adaptation on biological molecular sets.

Methods	NYC			CHI			SIP			SD			MUTAG	HIV	BBBP	Improvement (AIP)±std
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	ACC	ACC	ACC	/
BrainEVO	6.108	14.289	0.365	1.451	2.657	0.363	0.672	1.338	0.191	10.585	17.839	0.143	0.979	0.815	0.751	/
BrainEVO-HCL (intra-cluster)	7.007	16.661	0.374	1.696	3.239	0.380	0.699	1.389	0.204	12.071	19.932	0.183	0.957	0.799	0.742	5.80%±0.067
BrainEVO-HCL (inter-cluster)	6.763	15.982	0.367	1.667	3.146	0.370	0.691	1.376	0.203	11.962	19.334	0.177	0.964	0.804	0.741	4.54%±0.062
BrainEVO-Ela (static)	7.639	17.315	0.396	1.713	3.179	0.449	0.723	1.421	0.212	13.473	20.880	0.226	0.943	0.793	0.737	11.73%±0.11
BrainEVO-Ela (inverse proportion)	6.501	14.763	0.367	1.474	2.719	0.366	0.680	1.353	0.196	10.994	18.045	0.148	0.971	0.807	0.743	1.45%±0.009
BrainEVO-Ela (linear proportion)	6.577	14.952	0.368	1.488	2.801	0.368	0.684	1.369	0.199	11.046	18.243	0.150	0.971	0.806	0.744	1.96%±0.015
BrainEVO-MIL ($\rho = 0$)	6.626	15.005	0.384	1.641	2.846	0.392	0.720	1.378	0.215	13.123	21.285	0.179	0.921	0.787	0.729	8.07%±0.055
BrainEVO-MIL ($\kappa \to +\infty$)	7.104	16.719	0.376	1.702	3.331	0.379	0.717	1.365	0.214	13.207	21.772	0.183	0.914	0.784	0.72	7.71%± 0.063
BrainEVO-LCM	6.695	15.207	0.385	1.569	2.851	0.372	0.674	1.341	0.197	11.128	18.645	0.165	0.964	0.798	0.746	4.05%±0.04

all datasets, i.e.,

$$AIP = \frac{1}{N}(\frac{MAPE_V - MAPE_{BrainEVO}}{MAPE_V} + \frac{ACC_{BrainEVO} - ACC_V}{ACC_V}) \tag{25}$$

The variances of performance decreases are also incorporated into this metric to illustrate the robustness. This additional analysis provides a clearer and more direct illustration of the individual contribution and stability of each component, rather than relying only on the final performance numbers in the original tables. The added AIP *pm* std performance evaluations have been added into the most right column of Tab. 9.

From Tab. 9, we can easily decouple the importance of each module. First, the elastic module, i.e., dynamic adjustment of dropout factor and weight decay plays most important role and then memory-learning mutual interaction and finally the hierarchical curriculum learning.

3.5 Detailed evaluation of Learning Construct Memory (LCM)

Monitoring the activation frequencies of all neurons may cause many external computational costs. Actually, in our LCM, we would like to clarify that it does not monitor neuron activation frequencies in a layer-by-layer manner across the entire network. Instead, in our implementation, we only compute activation-frequency statistics for the *last neural activation layer*, since this layer is the most directly related to the final prediction output. We believe this strategy can well balance the effectiveness and efficiency. Therefore, the additional computation is limited where it only involves a lightweight vector-level aggregation over the existing activation tensor, without any extra backward passes or additional optimizer states.

To make this point more precise, we report the actual runtime with and without (-w/o LCM) on NYC and BBBP, where the running time of the full model against its ablated variants (-w/o LCM). The results have been demonstrated in Tab. 10. As illustrated, with LCM module, BrainEVO is more capable of countering data deficiency challenge, where the errors on NYC keeps much lower and accuracy on BBBP keeps higher as sparsity rate increasing. On computational cost aspect, our BrainEVO absolutely deserves LCM module as only approximately 0.5% overall running time can exchange the valuable robustness under extreme scenarios.

3.6 Detailed evaluation of BrainEVO's forgetting issue

To provide comprehensive evaluation on forgetting issue of our BrainEVO, we design two respective experiments.

Table 10: Detailed evaluations on LCM-based ablative variant on NYC and BBBP (MAE Performance/Runtime ratio)

Evaluation details	Full training set (Performance/Runtime ratio)	With 67% training set (Performance/Runtime ratio)	With 33% training set (Performance/Runtime ratio)
NYC, BrainEVO with LCM	6.108/0.38%	6.704/0.41%	7.439/0.42%
NYC, BrainEVO-w/o LCM	6.695/NULL	7.171/NULL	7.802/NULL
BBBP, BrainEVO with LCM	0.751/0.44%	0.729/0.46%	0.707/0.50%
BBBP, BrainEVO-w/o LCM	0.746/NULL	0.721/NULL	0.685/NULL

Table 11: Evaluation (MAE) of ablative variant of BrainEVO-LCM under catastrophic forgetting setting

Models	**NYC (Cross-source) (TaxiDrop/Crowd-in/Crowd-out)**	**NYC (Cross-temporal) Test on former three divisions**	**BBBP (Cross-source) Train on ++ then −− and test on ++**
BrainEVO with LCM	11.950/20.914/31.088	19.749/37.939/39.777	0.700
BrainEVO-w/o LCM	12.192/21.232/32.520	19.858/37.957/41.074	0.686
Backbone	15.847/27.939/40.649 (GWN)	23.923/42.659/46.747 (GWN)	0.626 (GIN)

The overall performance when testing on call-back datasets. Concretely, for evaluating the forgetting issue, we let the model call back to re-test on previous learned source domain when it has finished fine-tuning on last targeted domain. The detailed evaluation protocols are as follows.

- On NYC cross-source setting, BrainEVO has just fine-tuned on targeted TaxiPick, then it is required to make predictions on 10% of total samples on previous three domains, TaxiDrop, Crowd-in and Crowd-out, following 7:2:1 sample division paradigm.
- On cross-temporal setting, we have divided the training set into four division, [0-6 hour, 6-12 hour, 12-18 hour, 18-24 hour], and we lastly train on 18-24h and then retrieve the previous three divisions with 10% sampling rate for evaluating the forgetting issue.
- On molecule dataset, we let BrainEVO sequentially trains on ++, and −−, then we allow the testing on previous ++, where the sampling rate also keeps 10%.

Following above protocol, we report the performances of MAE metric in Tab. 12 for your convenience. As observed, we can see that our BrainEVO still performs robust against peer models when competing on the scenarios of catastrophic forgetting.

Evaluations with ablative LCM module under forgetting scenarios. We further report a revisiting evaluation on previously learned domains in Tab. 11, comparing the full model against the variant without LCM, in order to directly assess the contribution of LCM to mitigating forgetting. We can see that performances of ablative variants without LCM decrease ranging from 1.25% to 2.63%. In addition, our solution does not work in an independent manner, but it improves the robustness and alleviates forgetting in a jointed perspective with controllable elastic expansion, where the prominent performance decrease can be observed when comparing against Backbones (GWN on NYC and GIN on BBBP).

3.7 Detailed hyperparameter evaluation of BrainEVO

The major hyperparameters in BrainEVO can be categorized into four aspects.

- Dropout p_0 from $\{0.1, 0.3, 0.5, 0.7, 1\}$, controlling the scale of activated neurons for elastic model evolution.

Table 12: Evaluation on forgetting issue (MAE reported)

Models	NYC (Cross-source) (TaxiDrop/Crowd-in/Crowd-out)	NYC (Cross-temporal) Test on former three divisions	**Models**	BBBP (Cross-source) Train on ++ then −− and test on ++
GWN	15.847/27.939/40.649	23.923/42.659/46.747	GIN	0.626
CMuST	12.355/22.057/34.123	20.125/38.141/44.764	GAT	0.575
BrainEVO	11.950/20.914/31.088	19.749/37.939/39.777	BrainEVO	0.700

- Weight decay factor of L2 regularization, λ_0 from {0.01, 0.03, 0.05, 0.07, 0.1}, which adjusts the model complexity.
- Distance threshold κ from {1, 2, 3, 4, 5}, denotes the minimum distance across two domains.
- The low and high frequency threshold of neuron activations, $\epsilon_1 \in \{0.1, 0.2, 0.3, 0.4\}$ and high frequency $\epsilon_2 \in \{0.6, 0.7, 0.8, 0.9\}$ vary across two respective datasets on urban (NYC) and molecule (BBBP) sets.

The results can be found in Fig. 3-Fig. 6.

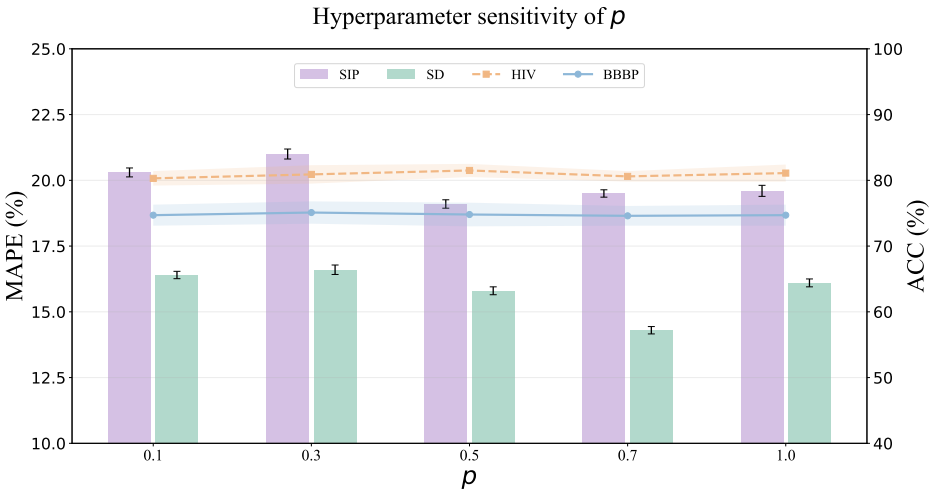


Figure 3: Updated parameter sensitivity analysis of initialized Dropout probability p_0.

Analysis on Dropout p_0. As observed, the performance variation patterns are similar across spatiotemporal datasets and molecule datasets (tasks), as ST datasets arrive best around $p_0 = 0.5$ while molecule datasets also behave in this way. This demonstrates 50% can be a general good probability for neuron activation at an initial stage, benefiting following evolving process.

Analysis on weight decay initialization λ_0. The λ_0 arrives best at 0.03 on spatiotemporal datasets, while 0.03 or 0.05 for molecules, where it reveals the consistency of general neural network behaviors in fitting diverse datasets.

Analysis on domain distance threshold κ**.** The parameter κ serves as a critical threshold governing the incorporation of new domains. An excessively small κ prohibits the admission of any new domain, thereby contradicting the principle of 'harmony with diversity for collective intelligence'. Conversely, an overly large κ admits noisy or irrelevant domains under overly relaxed fusion conditions, which ultimately degrades overall performance. Therefore, an appropriate value of κ must be carefully selected to strike a balance between extracting commonality and preserving distinctive domain-specific features, ensuring both

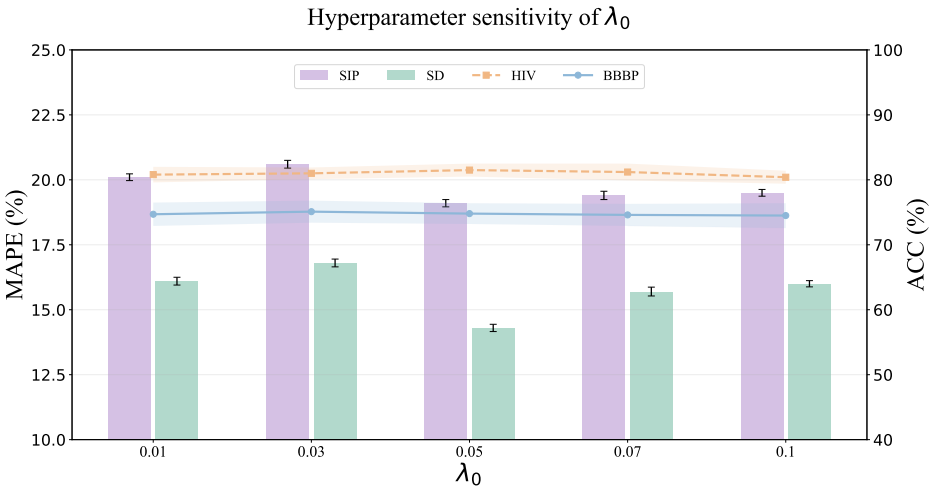


Figure 4: Updated parameter sensitivity analysis of initialized weight decay λ_0.

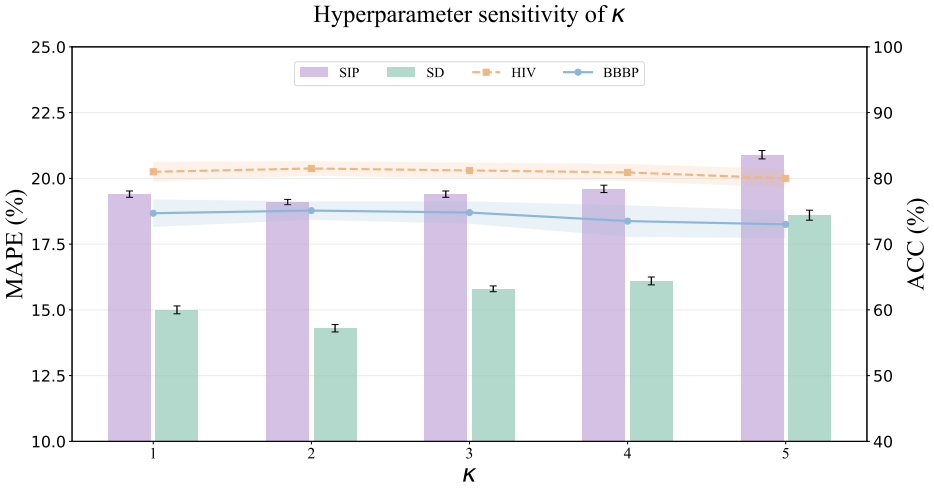


Figure 5: Updated parameter sensitivity analysis of domain distance threshold κ.

robustness and adaptability. Specifically, we observe that these four datasets consistently arrive $\kappa = 2$ for the best, which may indicate the similar model behaviors across different tasks.

Analysis on memory band (ϵ_1, ϵ_2). The ϵ_1 defines the low-frequency boundary of neuron activation, while ϵ_2 characterizes highly active and stable neurons during learning. For spatiotemporal datasets, we selected the hyperparameters yielding lower MAE, while for molecular property datasets, we selected those yielding higher Accuracy. These hyperparameters essentially characterize the model's tolerance range for moderately activated patterns. Due to differences in pattern stability and gradient consistency across data domains, neural networks may form memory selection bands of different widths, providing a computational analogy to activity-dependent strengthening and weakening associated with LTP/LTD.

Take activation-distribution-based quantiles as heuristic LCM thresholds. To reduce dataset-specific threshold tuning, we further adopt an activation-distribution-based heuristic. Given the empirical activation frequency distribution $\mathcal{V} = \{\nu_i\}$, where $\nu_i = N_{\text{activate},i}/N_{\text{total}}$, we define $\epsilon_1 = Q_{q_l}$ and $\epsilon_2 = Q_{q_u}$. We examine $\{q_l \in \{10\%, 15\%, 20\%, 25\%, 30\%\}$ and thus $q_u \in \{70\%, 75\%, 80\%, 85\%, 90\%\}$. The results can be

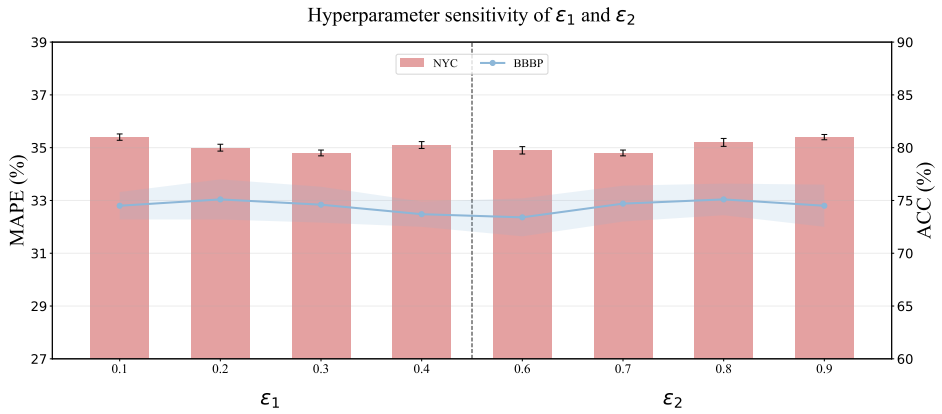


Figure 6: Updated parameter sensitivity analysis of ϵ.

Table 13: Performances across different Quantile-based thresholds for Learning Constructs Memory (LCM) on NYC and BBBP

NYC	$q_l(\%)$	10	15	**20**	25	30
	MAE	6.123	6.106	**6.097**	6.127	6.124
BBBP	$q_l(\%)$	10	15	20	**25**	30
	Acc	0.750	0.748	0.751	**0.754**	0.749

found in Table. 13 and we compare them against best one, finally recommend $\epsilon_1 = Q_{20\%}$, $\epsilon_2 = Q_{80\%}$ for spatiotemporal tasks and $\{\epsilon_1 = Q_{25\%}, \epsilon_2 = Q_{75\%}\}$ for molecular tasks. Generally, we can recommend $\{\epsilon_1 = Q_{20\%}, \epsilon_2 = Q_{80\%}\}$ for almost all datasets those are newly arrived in our system.

Specifically, spatiotemporal data usually exhibit stronger periodicity and continuity, leading to more stable patterns, more clustered embeddings, and more consistent gradient directions. As a result, the activation distribution is more concentrated, so the suppression boundary does not need to be very low and the enhancement boundary does not need to be very high, resulting in a relatively compact interval. In contrast, molecular datasets involve more diverse functional groups and combinatorial structures, leading to higher pattern diversity and weaker gradient consistency. Therefore, molecular tasks are closer to sparse causal pattern discovery, where only patterns that are highly stable, repeatedly observed across samples, and consistently beneficial to loss optimization are reinforced.

4 Reproducibility

All the resources, including datasets, code of models and evaluations as well as this Supplementary Information associated with our manuscript are available at https://github.com/Rodger-Lau/BrainEVO.

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